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BrightLearn

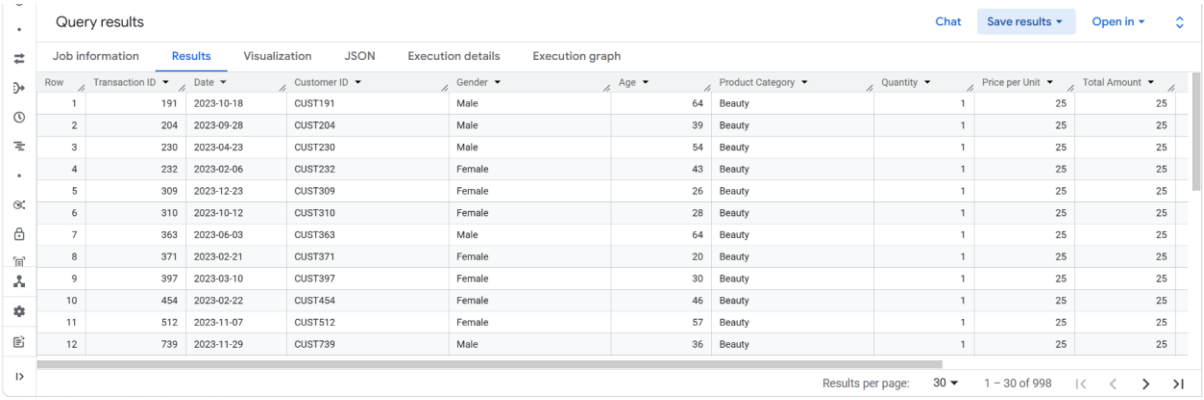
Practical Exercise Google BigQuery.

Dataset: Retail Sales Dataset.

Question 1: WHERE clause.

filter all transactions that occurred in the year 2023.

```
SELECT *
FROM `sales-projects-494801.Sales.Retailsales`
WHERE EXTRACT(YEAR FROM Date) = 2023;
```



Query results

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Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Price per Unit	Total Amount
1	191	2023-10-18	CUST191	Male	64	Beauty	1	25	25
2	204	2023-09-28	CUST204	Male	39	Beauty	1	25	25
3	230	2023-04-23	CUST230	Male	54	Beauty	1	25	25
4	232	2023-02-06	CUST232	Female	43	Beauty	1	25	25
5	309	2023-12-23	CUST309	Female	26	Beauty	1	25	25
6	310	2023-10-12	CUST310	Female	28	Beauty	1	25	25
7	363	2023-06-03	CUST363	Male	64	Beauty	1	25	25
8	371	2023-02-21	CUST371	Female	20	Beauty	1	25	25
9	397	2023-03-10	CUST397	Female	30	Beauty	1	25	25
10	454	2023-02-22	CUST454	Female	46	Beauty	1	25	25
11	512	2023-11-07	CUST512	Female	57	Beauty	1	25	25
12	739	2023-11-29	CUST739	Male	36	Beauty	1	25	25

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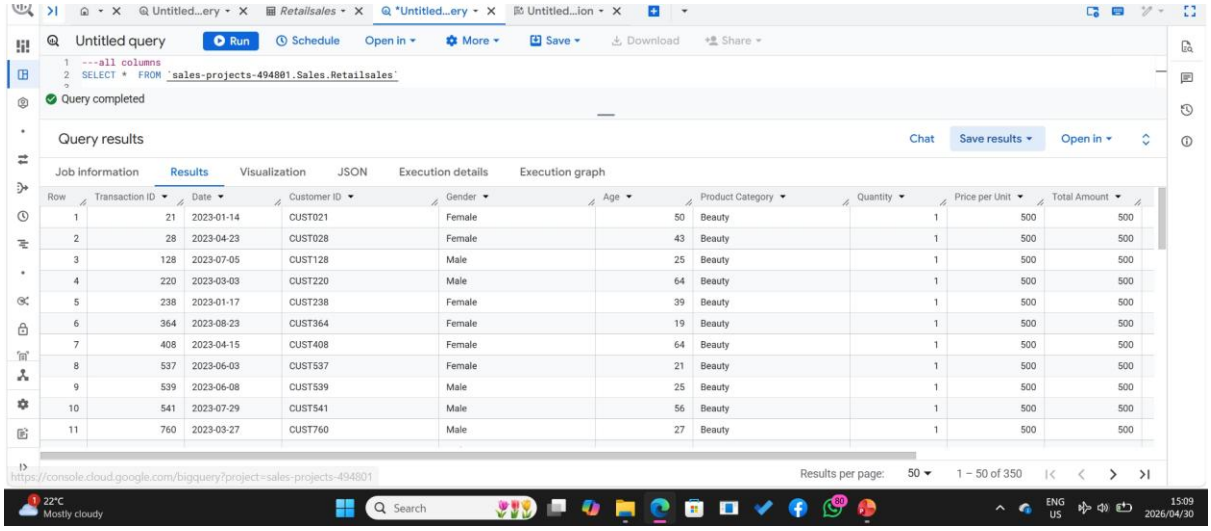
Question 2: Filtering + conditions

Display all transactions where the Total Amount is more than the average Total Amount of the entire dataset.

```
SELECT *
FROM `sales-projects-494801.Sales.Retailsales`
WHERE `Total Amount` > (SELECT AVG(`Total Amount`) FROM `sales-projects-494801.Sales.Retailsales`);
```



Untitled query | Run | Schedule | Open in | More | Save | Download | Share

```
---all columns
SELECT * FROM `sales-projects-494801.Sales.Retailsales`
```

Query completed

Query results

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Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Price per Unit	Total Amount
1	21	2023-01-14	CUST021	Female	50	Beauty	1	500	500
2	28	2023-04-23	CUST028	Female	43	Beauty	1	500	500
3	128	2023-07-05	CUST128	Male	25	Beauty	1	500	500
4	220	2023-03-03	CUST220	Male	64	Beauty	1	500	500
5	238	2023-01-17	CUST238	Female	39	Beauty	1	500	500
6	364	2023-08-23	CUST364	Female	19	Beauty	1	500	500
7	408	2023-04-15	CUST408	Female	64	Beauty	1	500	500
8	537	2023-06-03	CUST537	Female	21	Beauty	1	500	500
9	539	2023-06-08	CUST539	Male	25	Beauty	1	500	500
10	541	2023-07-29	CUST541	Male	56	Beauty	1	500	500
11	760	2023-03-27	CUST760	Male	27	Beauty	1	500	500

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Question 3: Aggregate functions

Calculate the total revenue (sum of Total Amount).

Expected Output: Total_Revenue.

```
SELECT SUM(`Total Amount`) AS Total_Revenue
FROM `sales-projects-494801.Sales.Retailsales`
```

Query results

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Total_Revenue
1	456000

Question 4: DISTINCT

Display all distinct Product Categories in the dataset.

Expected Output: Product_Category

```
SELECT DISTINCT `Product Category`
FROM `sales-projects-494801.Sales.Retailsales` ;
```

Query results

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Product Category
1	Beauty
2	Clothing
3	Electronics

Question 5: GROUP BY

For each Product Category, calculate the total quantity sold.

Expected Output: Product_Category, Total_Quantity

```
SELECT  `Product Category`,
        SUM(Quantity) AS Total_Quantity
  FROM `sales-projects-494801.Sales.Retailsales`
  GROUP BY `Product Category`;
```

Query results

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Product Category	Total_Quantity
1	Beauty	771
2	Clothing	894
3	Electronics	849

Question 6: CASE Statement

Create a column called Age_Group that classifies customers as 'Youth' (<30) 'Adult' (30–59) – 'Senior' (60+)

Expected Output: Customer_ID, Age, Age_Group

```
SELECT `Customer ID`,
       Age,
       CASE
           WHEN Age < 30 THEN 'Youth'
           WHEN Age BETWEEN 30 AND 59 THEN 'Adult'
           WHEN AGE > 60 THEN 'Senior'
           END AS Age_Group
 FROM `sales-projects-494801.Sales.Retailsales`;
```

Query results

Chat | Save results | Open in

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Customer ID	Age	Age_Group
1	CUST191	64	Senior
2	CUST204	39	Adult
3	CUST230	54	Adult
4	CUST232	43	Adult
5	CUST309	26	Youth
6	CUST310	28	Youth
7	CUST363	64	Senior
8	CUST371	20	Youth
9	CUST397	30	Adult
10	CUST454	46	Adult
11	CUST512	57	Adult
12	CUST739	36	Adult

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Question 7: Conditional Aggregation

For each Gender, count how many high-value transactions occurred (where Total Amount > 500).

Expected Output: Gender, High_Value_Transactions

```
SELECT Gender,
       COUNT(CASE WHEN `Total Amount` > 500 THEN 1 END) AS
High_Value_Transactions
FROM `sales-projects-494801.Sales.Retailsales`
```

```
GROUP BY Gender;
```

Query results

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Gender	High_Value_Tran...
1	Male	144
2	Female	155

Question 8: HAVING Clause

For each Product Category, show only those categories where the total revenue exceeds 5,000.

Expected Output: Product_Category, Total_Revenue

```
SELECT  `Product Category`,
        SUM(`Total Amount`) AS Total_Revenue
  FROM `sales-projects-494801.Sales.Retailsales`
  GROUP BY `Product Category`
  HAVING SUM(`Total Amount`) > 5000;
```

Row	Product Category	Total_Revenue
1	Beauty	143515
2	Clothing	155580
3	Electronics	156905

Question 9: Calculated Fields

Display a new column called Unit_Cost_Category that labels a transaction as:

'Cheap' if Price per Unit < 50

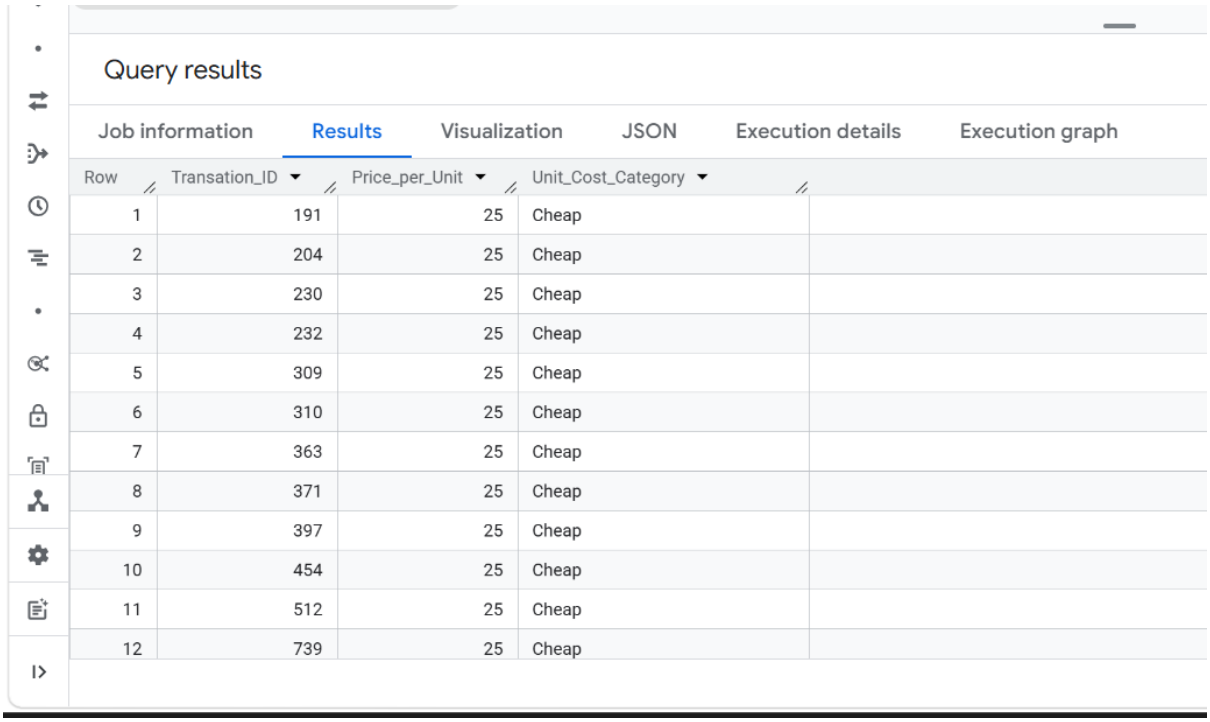
'Moderate' if Price per Unit between 50 and 200

'Expensive' if Price per Unit > 200

Expected Output: Transaction_ID, Price_per_Unit, Unit_Cost_Category

```
SELECT `Transaction ID` AS Transation_ID,
`Price per Unit` AS Price_per_Unit,
       CASE
           WHEN `Price per Unit` < 50 THEN 'Cheap'
           WHEN `Price per Unit` BETWEEN 50 AND 200 THEN 'Moderate'
           WHEN `Price per Unit` > 200 THEN 'Expensive'
```

```
            ELSE 'Unknown'
        END AS Unit_Cost_Category
FROM `sales-projects-494801.Sales.Retailsales`;
```



Query results

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Transation_ID	Price_per_Unit	Unit_Cost_Category
1	191	25	Cheap
2	204	25	Cheap
3	230	25	Cheap
4	232	25	Cheap
5	309	25	Cheap
6	310	25	Cheap
7	363	25	Cheap
8	371	25	Cheap
9	397	25	Cheap
10	454	25	Cheap
11	512	25	Cheap
12	739	25	Cheap

Question 10: Combining WHERE + CASE

Display all transactions from customers aged 40 or older and add a column Spending_Level showing:

High' if Total Amount > 1000

'Low' otherwise

Expected Output: Customer_ID, Age, Total_Amount, Spending_Level

```
SELECT `Customer ID`,
       Age,
       `Total Amount`,
       CASE
           WHEN `Total Amount` > 1000 THEN 'High'
           ELSE 'Low'
       END AS Spending_Level
FROM `sales-projects-494801.Sales.Retailsales`
WHERE Age >= 40;
```

Untitled query

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This query will process 24.42 KB when run.

Using on-demand processing quota

Query results

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Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Customer ID	Age	Total Amount	Spending_Level
1	CUST191	64	25	Low
2	CUST230	54	25	Low
3	CUST232	43	25	Low
4	CUST363	64	25	Low
5	CUST454	46	25	Low
6	CUST512	57	25	Low
7	CUST791	51	25	Low
8	CUST825	46	25	Low
9	CUST855	54	25	Low
10	CUST967	62	25	Low
11	CUST006	45	30	Low
12	CUST057	63	30	Low
13	CUST344	42	30	Low

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